variables concerning the existence of higher-order derivatives. In real analysis, if a function f possesses, say, a first derivative, there is no guarantee that f possesses any other higher derivatives. For example, on the interval $(-1,\ 1)$, $f(x) = x^{3/2}$ is differentiable at $x = 0$, but $f'(x) = \frac{3}{2}x^{1/2}$ is not differentiable at $x = 0$. In complex analysis, if a function f is *analytic* in a domain D then, by assumption, f possesses a derivative at each point in D. We will see in Section 5.5 that this fact alone guarantees that f possesses higher-order derivatives at all points in D. Indeed, an analytic function f is infinitely differentiable in D.

(*v*) The definition of "analytic at a point a" in real analysis differs from the usual definition of that concept in complex analysis (Definition 3.2.2). In real analysis, analyticity of a function is defined in terms of power series: A function $y = f(x)$ is analytic at a point a if f has a Taylor series at a that represents f in some neighborhood of a. In view of *Remark* (*iv*), why are these two definitions not really that different?

(*vi*) As in real calculus, it may be necessary to apply L'Hôpital's rule several times in succession to calculate a limit. In other words, if $f(z_0),\ g(z_0),\ f'(z_0)$, and $g'(z_0)$ are all zero, the limit $\lim\limits_{z\to z_0} f(z)/g\,(z)$ may still exist. In general, if f, g, and their first $n-1$ derivatives are zero at z_0 and $g^{(n)}(z_0) \neq 0$, then

$$\lim_{z\to z_0} \frac{f(z)}{g(z)} = \frac{f^{(n)}(z_0)}{g^{(n)}(z_0)}.$$

EXERCISES 3.2

Answers to selected odd-numbered problems begin on page ANS-11.

In Problems 1–6, use (1) of Definition 3.2.1 to find $f'(z)$ for the given function.

1. $f(z) = 9iz + 2 - 3i$

2. $f(z) = 15z^2 - 4z + 1 - 3i$

3. $f(z) = iz^3 - 7z^2$

4. $f(z) = \dfrac{1}{z}$

5. $f(z) = z - \dfrac{1}{z}$

6. $f(z) = -z^{-2}$

In Problems 7–10, use the alternative definition (12) to find $f'(z)$ for the given function.

7. $f(z) = 5z^2 - 10z + 8$

8. $f(z) = z^3$

9. $f(z) = z^4 - z^2$

10. $f(z) = \dfrac{1}{2iz}$

In Problems 11–18, use the rules of differentiation to find $f'(z)$ for the given function.

11. $f(z) = (2-i)z^5 + iz^4 - 3z^2 + i^6$

12. $f(z) = 5(iz)^3 - 10z^2 + 3 - 4i$

13. $f(z) = (z^6 - 1)(z^2 - z + 1 - 5i)$

14. $f(z) = (z^2 + 2z - 7i)^2(z^4 - 4iz)^3$

15. $f(z) = \dfrac{iz^2 - 2z}{3z + 1 - i}$

16. $f(z) = -5iz^2 + \dfrac{2+i}{z^2}$

17. $f(z) = \left(z^4 - 2iz^2 + z\right)^{10}$

18. $f(z) = \left(\dfrac{(4+2i)z}{(2-i)z^2 + 9i}\right)^3$

19. The function $f(z) = |z|^2$ is continuous at the origin.

(**a**) Show that f *is* differentiable at the origin.

(**b**) Show that f *is not* differentiable at any point $z \neq 0$.

20. Show that the function

$$f(z) = \begin{cases} 0, & z = 0 \\ \dfrac{x^3 - y^3}{x^2 + y^2} + i\dfrac{x^3 + y^3}{x^2 + y^2}, & z \neq 0 \end{cases}$$

is not differentiable at $z = 0$ by letting $\Delta z \to 0$ first along the x-axis and then along the line $y = x$.

In Problems 21 and 22, proceed as in Example 3 to show that the given function is nowhere differentiable.

21. $f(z) = \bar{z}$

22. $f(z) = |z|$

In Problems 23–26, use L'Hôpital's rule to compute the given limit.

23. $\lim\limits_{z \to i} \dfrac{z^7 + i}{z^{14} + 1}$

24. $\lim\limits_{z \to \sqrt{2}+\sqrt{2}i} \dfrac{z^4 + 16}{z^2 - 2\sqrt{2}z + 4}$

25. $\lim\limits_{z \to 1+i} \dfrac{z^5 + 4z}{z^2 - 2z + 2}$

26. $\lim\limits_{z \to \sqrt{2}i} z\dfrac{z^3 + 5z^2 + 2z + 10}{z^5 + 2z^3}$

In Problems 27–30, determine the points at which the given function is not analytic.

27. $f(z) = \dfrac{iz^2 - 2z}{3z + 1 - i}$

28. $f(z) = -5iz^2 + \dfrac{2 + i}{z^2}$

29. $f(z) = \left(z^4 - 2iz^2 + z\right)^{10}$

30. $f(z) = \left(\dfrac{(2 - i)z + 1}{z^2 + 9i}\right)^3$

Focus on Concepts

31. Use Definition 3.2.1 and Theorem 3.1.2 to prove the constant rule $\dfrac{d}{dz}cf(z) = cf'(z)$.

32. Use Definition 3.2.1 and Theorem 3.1.2 to prove the sum rule $\dfrac{d}{dz}[f(z) + g(z)] = f'(z) + g'(z)$.

33. In this problem you are guided through the start of the proof of the proposition:

If functions f and g are analytic at a point z_0 and $f(z_0) = 0,\ g(z_0) = 0$, but $g'(z_0) \neq 0$, then $\lim\limits_{z \to z_0} \dfrac{f(z)}{g(z)} = \dfrac{f'(z_0)}{g'(z_0)}$.

Proof We begin with the hypothesis that f and g are analytic at a point z_0. Analyticity at z_0 implies f and g are differentiable at z_0. Hence from (12) both limits,

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \quad \text{and} \quad g'(z_0) = \lim_{z \to z_0} \frac{g(z) - g(z_0)}{z - z_0}$$

exist. But since $f(z_0) = 0,\ g(z_0) = 0$, the foregoing limits are the same as

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z)}{z - z_0} \quad \text{and} \quad g'(z_0) = \lim_{z \to z_0} \frac{g(z)}{z - z_0}.$$

Now examine $\lim\limits_{z \to z_0} \dfrac{f(z)}{g(z)}$ and finish the proof.

34. In this problem you are guided through the start of the proof of the product rule.

Proof We begin with the hypothesis that f and g are differentiable at a point z; that is, each of the following limits exist:

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \quad \text{and} \quad g'(z) = \lim_{\Delta z \to 0} \frac{g(z + \Delta z) - g(z)}{\Delta z}$$

(**a**) Justify the equality

$$\frac{d}{dz}[f(z)g(z)] = \lim_{\Delta z \to 0} \frac{f(z+\Delta z)g(z+\Delta z) - f(z)g(z)}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \left[\frac{f(z+\Delta z) - f(z)}{\Delta z} g(z+\Delta z) + f(z) \frac{g(z+\Delta z) - g(z)}{\Delta z} \right].$$

(**b**) Use Definition 3.1.2 to justify $\lim_{\Delta z \to 0} g(z+\Delta z) = g(z)$.

(**c**) Use Theorems 3.1.2(*ii*) and 3.1.2(*iii*) to finish the proof.

35. In this problem you will prove that $f(z) = \overline{z}$ is nowhere differentiable using the polar form $\Delta z = r(\cos\theta + i\sin\theta)$.

(**a**) What must r approach if $\Delta z \to 0$? Must θ approach a specific value?

(**b**) If $f(z) = \overline{z}$, show that

$$\lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = \frac{\cos\theta - i\sin\theta}{\cos\theta + i\sin\theta}.$$

(**c**) Explain succinctly why the result in part (b) shows that f is nowhere differentiable.

36. In this problem you will prove that $f(z) = |z|$ is nowhere differentiable using the alternative definition (12) and polar forms of z and z_0.

(**a**) Let $z_0 = r(\cos\theta + i\sin\theta)$ and $z = (r+\Delta r)(\cos\theta + i\sin\theta)$. Along this path determine $\lim_{z \to z_0} \dfrac{f(z) - f(z_0)}{z - z_0}$.

(**b**) Let $z_0 = r(\cos\theta + i\sin\theta)$ and $z = r\,[\cos(\theta + \Delta\theta) + i\sin(\theta + \Delta\theta)]$. Along this path determine $\lim_{z \to z_0} \dfrac{f(z) - f(z_0)}{z - z_0}$.

(**c**) Explain succinctly why the results in parts (a) and (b) show that f is nowhere differentiable.

37. Suppose $f'(z)$ exists at a point z. Is $f'(z)$ continuous at z?

38. (**a**) Let $f(z) = z^2$. Write down the real and imaginary parts of f and f'. What do you observe?

(**b**) Repeat part (a) for $f(z) = 3iz + 2$.

(**c**) Make a conjecture about the relationship between real and imaginary parts of f versus f'.

3.3 Cauchy-Riemann Equations

INTRODUCTION In the preceding section we saw that a function f of a complex variable z is *analytic at a point* z when f is differentiable at z and differentiable at every point in some neighborhood of z. This requirement is more stringent than simply *differentiability at a point* because a complex function can be differentiable at a point z but yet be differentiable nowhere else. A function f is *analytic in a domain* D if f is differentiable at all points in D. We shall now develop a test for analyticity of a complex function $f(z) = u(x, y) + iv(x, y)$ that is based on partial derivatives of its real and imaginary parts u and v.

A Necessary Condition for Analyticity

A review of partial derivatives can be found in Section 3.3 of the student study guide. ▶

In the next theorem we see that if a function $f(z) = u(x, y) + iv(x, y)$ is differentiable at a point z, then the functions u and v must satisfy a pair of equations that relate their first-order partial derivatives.